

Transforming California Independent Medical Reviews Data into Actionable Intelligence

An End-to-End Data Warehouse and AI Implementation on Snowflake

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The Project Mandate: From Raw Files to Refined Insights



The Challenge

- To ingest, model, and analyze a complex dataset of Independent Medical Reviews (IMR) - 19K rows, 11 columns.
- The goal is to build a robust, scalable, and automated data platform that supports business intelligence and advanced analytics.

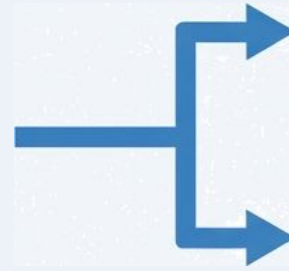


Our Solution: A Medallion Architecture

- We designed and deployed an end-to-end data pipeline on the Snowflake Data Cloud.
- The core of our solution is a Medallion Architecture (Bronze, Silver, Gold layers) to progressively refine data.
- This is augmented with Snowflake Cortex AI to unlock insights from both structured and unstructured data.



Raw
Files

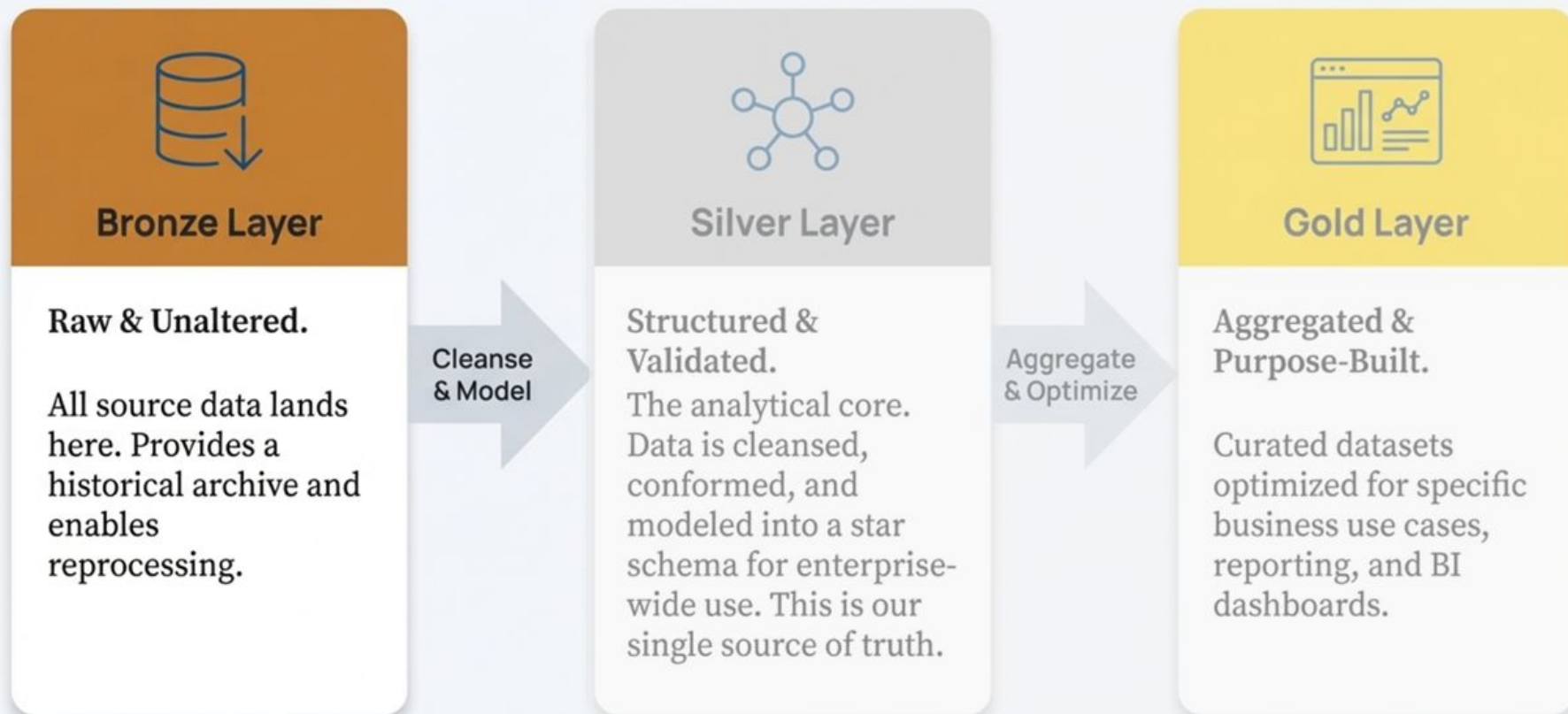


BI &
Analytics



AI-Driven
Insights

Our Architectural Blueprint: The Data's Journey



Ingesting the Raw Material into the Bronze Layer

Objective

To load the complete, unaltered IMR dataset into a dedicated BRONZE schema, creating a persistent staging area.

Execution

A BRONZE schema was created within our project database DB TEAM QUERYQUEST. The raw IMR dataset was loaded into a single table BRONZE.IMR_RAW

```
SQL> as create_data_table •
1 CREATE OR REPLACE FILE FORMAT my_csv_format
2 TYPE = 'CSV'
3 SKIP_HEADER = 1
4 FIELD_OPTIONALLY_ENCLOSED_BY = ''
5 NULL_IF = ('NULL', 'null', '')
6 EMPTY_FIELD_AS_NULL = TRUE;
7
8 CREATE OR REPLACE TABLE IMR_RAW AS
9 SELECT
10 $1 AS Reference_ID,
11 $2 AS Report_Year,
12 $3 AS Diagnosis_Category,
13 $4 AS Diagnosis_Sub_Category,
14 $5 AS Treatment_Category,
15 $6 AS Treatment_Sub_Category,
16 $7 AS Determination,
17 $8 AS Review_Type,
18 $9 AS Age_Range,
19 $10 AS Patient_Gender,
20 $11 AS Findings
21 FROM @PROJECT_STAGE/CaI_Independent_Medical_Reviews.csv
22 (FILE_FORMAT => my_csv_format);
23 SELECT * FROM IMR_RAW;
24
```

REFERENCE_ID	REPORT_YEAR	DIAGNOSIS_CATEGORY	DIAGNOSIS_SUB_CATEGORY	TREATMENT_CATEGORY	TREATMENT_SUB_CATEGORY	DETERMINATION	REVIEW_TYPE	AGE_RANGE	PATIENT
EI05-4327	2005	Central Nervous System/ Neu	Other	Neurosugery	Other	Upheld Decision of Health Plan	Experimental/Investigational	31-40	Male
MN05-4326	2005	Orthopedic/ Musculoskeletal	Joint Problem	Durable Medical Equipment	Orthotics	Upheld Decision of Health Plan	Medical Necessity	31-40	Female
MN05-4325	2005	Central Nervous System/ Neu	Headache/Migraine	Alternative Treatment	Acupuncture	Overturned Decision of Health Plan	Medical Necessity	0-10	Male
MN05-4324	2005	Mental	Substance Abuse/ Addiction	Mental Health Treatment	Drug Rehabilitation Program	Overturned Decision of Health Plan	Medical Necessity	11-20	Male
MN05-4323	2005	Respiratory System	COPD - Chronic Obstructive L	Durable Medical Equipment	Other	Upheld Decision of Health Plan	Medical Necessity	41-50	Female
MN05-4322	2005	Morbid Obesity	Post Surgical Complication	Reconstructive/Plastic Surgery	Other	Upheld Decision of Health Plan	Medical Necessity	31-40	Female
MN05-4321	2005	Mental	Alcohol Abuse/ Addiction	Rehabilitation Services - Skilled Nursing Facility - Inpatient	Inpatient Admission	Overturned Decision of Health Plan	Medical Necessity	51-64	Female
MN05-4320	2005	Morbid Obesity	Abdominal Pannus	Reconstructive/Plastic Surgery	Panniculectomy	Upheld Decision of Health Plan	Medical Necessity	41-50	Female
EI05-4319	2005	Dental	None	Pharmacy/Prescription Drugs	Botox Injection	Overturned Decision of Health Plan	Experimental/Investigational	41-50	Female
MN05-4318	2005	Post Surgical Complication	Other	Reconstructive/Plastic Surgery	Other	Upheld Decision of Health Plan	Medical Necessity	41-50	Female

AI Powered Enrichment: Enhancing Raw Data with Cortex

Objective

Transform raw Independent Medical Review (IMR) data into an enriched dataset using Snowflake Cortex AI.

Execution

Sentiment Analysis: Extended the schema to calculate sentiment scores on clinical Findings using SNOWFLAKE.CORTEX.SENTIMENT. This quantifies the emotional tone of the medical review text.

Automated Summarization: Generated concise summaries of complex medical narratives using SNOWFLAKE.CORTEX.SUMMARIZE, making long-form text immediately consumable for analysis.

```
ALTER TABLE IMR_RAW ADD COLUMN SENTIMENT FLOAT;  
ALTER TABLE IMR_RAW ADD COLUMN SUMMARY STRING;  
  
UPDATE IMR_RAW  
SET  
    SENTIMENT = SNOWFLAKE.CORTEX.SENTIMENT(Findings),  
    SUMMARY = SNOWFLAKE.CORTEX.SUMMARIZE(Findings);  
  
SELECT Reference_ID, Findings, SENTIMENT, SUMMARY  
FROM IMR_RAW  
LIMIT 10;
```

	REFERENCE_ID	FINDINGS	SENTIMENT	SUMMARY
0	MN16-22639	Nature of Statutory Criteria/Case Summary: An enrollee has requested Har	0.0391	The text describes a case where a patient with hepatitis C virus genotype
1	MN16-22638	Nature of Statutory Criteria/Case Summary: An enrollee has requested re	-0.1523	The enrollee requested residential treatment for her medical condition, but
2	MN16-22637	Nature of Statutory Criteria/Case Summary: The parent of an enrollee has	-0.0586	The parent requested speech therapy for their child with autism spectrum
3	EI16-22636	Nature of Statutory Criteria/Case Summary: An enrollee has requested bre	0.3359	Three board-certified diagnostic radiologists, all experts in the enrollee's n
4	EI06-5319	Physician 1: The patient is a 62-year-old male who is reported to have sma	-0.5566	In the first case, a 62-year-old male with small vessel disease and atypical
5	EI16-22635	Nature of Statutory Criteria/Case Summary: An enrollee has requested ad	-0.0937	Three medical experts determined that advanced lipoprotein testing was n
6	EI16-22634	Nature of Statutory Criteria/Case Summary: An enrollee has requested bre	0.2813	Two reviewers, who are board certified diagnostic radiologists and experts
7	EI16-22633	Nature of Statutory Criteria/Case Summary: An enrollee has requested bre	0.5703	Two radiologists concluded that breast tomosynthesis is more beneficial t
8	MN16-22632	Nature of Statutory Criteria/Case Summary: The parent of an enrollee has	0.1563	The parent requested occupational therapy for a medical condition, but thi
9	EI16-22158	Nature of Statutory Criteria/Case Summary: The parent of an enrollee has	-0.3203	Three expert reviewers determined that blood allergy testing, which the p

Intelligent Search Service: Cortex Implementation

Objective

Enable semantic search and information retrieval over medical findings using Cortex Search Services.

Execution

Deployed IMR_SEARCH_SERVICE on the unstructured Findings text column.

Configured Diagnosis_Category and Report_Year as key attributes to allow for precise filtering and faceted search.

Demonstrated advanced querying by combining semantic search with structured filters (e.g., finding "surgery complications" specifically for the year 2016).

```
WITH search_results AS (
  SELECT PARSE_JSON(
    SNOWFLAKE.CORTX.SEARCH_PREVIEW(
      'IMR_SEARCH_SERVICE',
      {
        "query": "surgery complications",
        "columns": [
          "Findings",
          "Diagnosis_Category",
          "Report_Year"
        ],
        "filter": { "@eq": { "Report_Year": "2016" } },
        "limit": 10
      }
    ) AS json_data
  ) AS json_data
)
SELECT
  r.value:"Diagnosis_Category"::STRING AS Diagnosis_Category,
  r.value:"Report_Year"::STRING AS Report_Year,
  r.value:"Findings"::STRING AS Findings,
  r.value:"@scores":"cosine_similarity"::FLOAT AS cosine_similarity,
  r.value:"@scores":"text_match"::FLOAT AS text_match
FROM search_results,
  LATERAL FLATTEN(input => json_data['results']) r;
```

DIAGNOSIS_CATEGORY	REPORT_YEAR	FINDINGS	COSINE_SIMILARITY	TEXT_MATCH
Digestive System/ Gastrointestinal	2016	Nature of Statutory Criteria/Case Summary: An enrollee has requested em	0.4112	0.022
Orthopedic/ Musculoskeletal	2016	Nature of Statutory Criteria/Case Summary: An enrollee has requested su	0.3934	0.0104
Cancer	2016	Nature of Statutory Criteria/Case Summary: An enrollee has requested pr	0.4237	0.0078
Respiratory System	2016	Nature of Statutory Criteria/Case Summary: An enrollee has requested bar	0.389	0.0207
Morbid Obesity	2016	Nature of Statutory Criteria/Case Summary: An enrollee has requested the	0.3936	0.0207
Morbid Obesity	2016	Nature of Statutory Criteria/Case Summary: An enrollee has requested a n	0.3762	0.0207
Cardiac/Circulatory	2016	Nature of Statutory Criteria/Case Summary: The parent of an enrollee has	0.3675	0.0078
Skin	2016	Nature of Statutory Criteria/Case Summary: An enrollee has requested re	0.3773	0.0083
Orthopedic/ Musculoskeletal	2016	Nature of Statutory Criteria/Case Summary: An enrollee has requested hip	0.3711	0.0078
Orthopedic/ Musculoskeletal	2016	Nature of Statutory Criteria/Case Summary: An enrollee has requested hip	0.382	0

Intelligent Search Service: Cortex Implementation

Findings that had heart failures

Records containing both 'neurology' and 'surgery' for 2016 report year

```
1  -- Test Query 1: Concept search ["heart failure"]
2  WITH search_results AS (
3    SELECT PARSE_JSON(
4      SNOWFLAKE.CORTX.SEARCH_PREVIEW(
5        'IMR_SEARCH_SERVICE',
6        '{
7          "query": "heart failure",
8          "columns": [
9            "Findings",
10           "Diagnosis_Category",
11           "Report_Year"
12         ],
13         "limit": 10
14       }'
15     ) AS json_data
16   )
17   SELECT
18     r.value:"Diagnosis_Category"::STRING AS Diagnosis_Category,
19     r.value:"Report_Year"::STRING AS Report_Year,
20     r.value:"Findings"::STRING AS Findings,
21     r.value:"@scores"::cosine_similarity::FLOAT AS cosine_similarity,
22     r.value:"@scores"::text_match::FLOAT AS text_match
23   FROM search_results,
24   LATERAL FLATTEN(input => json_data['results']) r;
```

	DIAGNOSIS_CATEGORY	REPORT_YEAR	FINDINGS	COSINE_
0	Cardiac/Circulatory	2006	Physician 1: The patient is a 47-year-old male with ischemic cardiomyopathy	
1	Cardiac/Circulatory	2007	A 75-year-old male enrollee has requested authorization and coverage for	
2	Cardiac/Circulatory	2015	A 65-year-old male enrollee has requested the wearable cardioverter defibr	
3	Cardiac/Circulatory	2014	A 66-year-old male enrollee has requested a cardiac positron emission tom	
4	Cardiac/Circulatory	2013	A 55-year-old female enrollee has requested Zoll LifeVest for treatment of	
5	Cardiac/Circulatory	2014	A 49-year-old male enrollee has requested wearable cardioverter defibrilla	
6	Cardiac/Circulatory	2011	A 37-year-old female enrollee has requested Bystolic for treatment of her	
7	Cardiac/Circulatory	2012	A 44-year-old male has requested Xarelto (rivaroxaban) for treatment of hi	
8	Cardiac/Circulatory	2015	Nature of Statutory Criteria/Case Summary: An enrollee has requested ca	
9	Cardiac/Circulatory	2005	The patient is a 78-year-old male who underwent cardiac transplant for cl	

```
1  -- Test Query 3: Complex Search (Keywords + Year Filter)
2  -- Demonstrates searching for records containing BOTH "neurology" and "surgery"
3  -- while strictly filtering for the Report Year 2016.
4
5  USE SCHEMA BRONZE;
6  WITH search_results AS (
7    SELECT PARSE_JSON(
8      SNOWFLAKE.CORTX.SEARCH_PREVIEW(
9        'IMR_SEARCH_SERVICE',
10       '{
11         "query": "neurology AND surgery",
12         "columns": [
13           "Findings",
14           "Diagnosis_Category",
15           "Report_Year"
16         ],
17         "filter": {
18           "eq": { "Report_Year": "2016" }
19         },
20         "limit": 10
21       }'
22     ) AS json_data
23   )
24   SELECT
25     r.value:"Diagnosis_Category"::STRING AS Diagnosis_Category,
26     r.value:"Report_Year"::INT AS Report_Year,
27     r.value:"Findings"::STRING AS Findings,
28     r.value:"@scores"::cosine_similarity::FLOAT AS cosine_similarity,
29     r.value:"@scores"::text_match::FLOAT AS text_match
30   FROM search_results,
31   LATERAL FLATTEN(input => json_data['results']) r
32   ORDER BY cosine_similarity DESC;
```

	DIAGNOSIS_CATEGORY	REPORT_YEAR	FINDINGS
0	Central Nervous System/ Neuromuscular	2,016	Nature of Statutory Criteria/Case Summary: An enrollee has requ
1	Central Nervous System/ Neuromuscular	2,016	Nature of Statutory Criteria/Case Summary: An enrollee has reques
2	Orthopedic/ Musculoskeletal	2,016	Nature of Statutory Criteria/Case Summary: An enrollee has reques
3	Orthopedic/ Musculoskeletal	2,016	Nature of Statutory Criteria/Case Summary: An enrollee has reques
4	Central Nervous System/ Neuromuscular	2,016	Nature of Statutory Criteria/Case Summary: An enrollee has reques

The Core Process: Silver Layer



Bronze Layer

Raw & Unaltered.

All source data lands here. Provides a historical archive and enables reprocessing.

Cleanse
& Model



Silver Layer

Structured & Validated.

The analytical core. Data is cleansed, conformed, and modeled into a star schema for enterprise-wide use. This is our single source of truth.

Aggregate
& Optimize



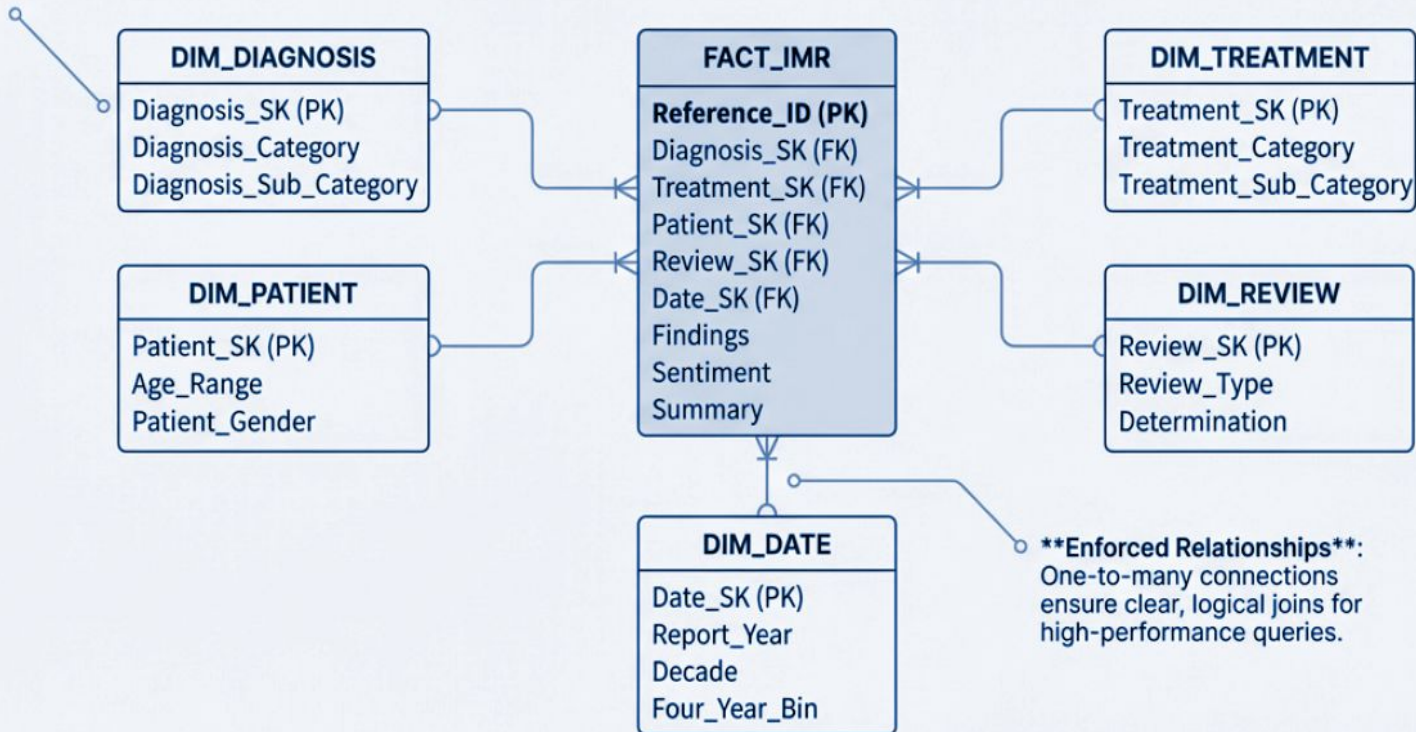
Gold Layer

Aggregated & Purpose-Built.

Curated datasets optimized for specific business use cases, reporting, and BI dashboards.

The Blueprint: A Star Schema for Performance and Governance

****Surrogate Keys**:** Generated using Snowflake Sequences to uniquely identify dimension records and decouple from source data.



Building the Silver Layer Schema: Dimension and Fact Integration

This stage defines structured schema objects and enforces relationships between dimensions and the fact table. Each dimension applies unique business rules, while joins and COALESCE() ensure null-safe mappings from the Bronze layer.

```
-- 3. DIM_PATIENT
CREATE OR REPLACE TABLE DIM_PATIENT (
  Patient_SK INT PRIMARY KEY,
  Age_Range VARCHAR(50) NOT NULL,
  Patient_Gender VARCHAR(50) NOT NULL,
  CONSTRAINT UK_PATIENT UNIQUE (Age_Range, Patient_Gender)
);
```

Defines Patient dimension using surrogate key

Ensures uniqueness using Age and Gender

Joins Diagnosis dimension using null-safe matching

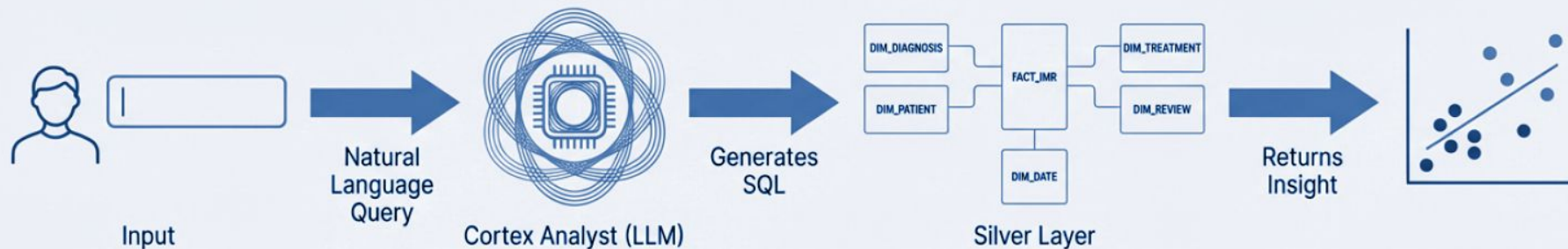
Joins Treatment dimension ensuring clean mapping

```
INSERT INTO FACT_IMR (
  Reference_ID, Diagnosis_SK, Treatment_SK, Patient_SK, Review_SK, Date_SK,
  Findings, Sentiment, Summary
)
SELECT
  b.Reference_ID, dd.Diagnosis_SK, dt.Treatment_SK, dp.Patient_SK, dr.Review_SK, ddate.Date_SK,
  b.Findings, b.Sentiment, b.Summary
FROM BRONZE.IMR_RAW b

-- DIM_DIAGNOSIS:
INNER JOIN DIM_DIAGNOSIS dd
  ON COALESCE(TRIM(b.Diagnosis_Category), 'Unspecified') = dd.Diagnosis_Category
  AND COALESCE(TRIM(b.Diagnosis_Sub_Category), 'Unspecified') = dd.Diagnosis_Sub_Category

-- DIM_TREATMENT:
INNER JOIN DIM_TREATMENT dt
  ON COALESCE(TRIM(b.Treatment_Category), 'Unspecified') = dt.Treatment_Category
  AND COALESCE(TRIM(b.Treatment_Sub_Category), 'Unspecified') = dt.Treatment_Sub_Category
```

Natural Language Querying with Cortex Analyst



Compare the average sentiment between male and female patients

PATIENT_GENDER	AVG_SENTIMENT
Male	0.02809551782
Female	-0.005208176727

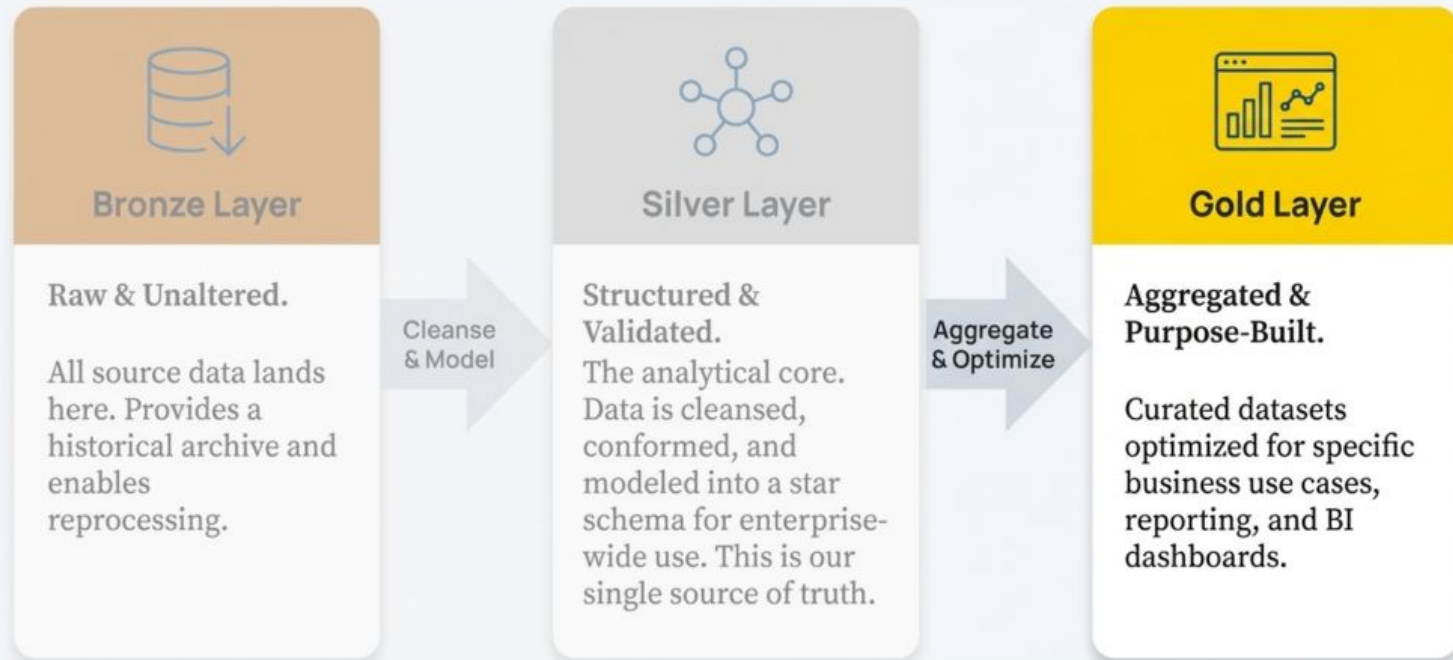
Which treatment category has the highest number of reviews for male patients in the 21-30 age range?

TREATMENT_CATEGORY	NUMBER_OF_REVIEWS
Mental Health Treatment	142

Identify outlier records where the sentiment score is highly positive, but the determination is overturned.

REFERENCE_ID	SENTIMENT	DETERMINATION	SUMMARY
EI10-11295	0.8828125	Overturned Decision of Health	A 50-year-old male with bipolar disorder, ADHD, and obstructive sleep apnea re
MN07-7016	0.8828125	Overturned Decision of Health	A 60-year-old woman requested Xyrem for insomnia treatment. The physician fo
MN09-9745	0.8828125	Overturned Decision of Health	A 48-year-old woman with hereditary peripheral neuropathy has been prescriber
EI11-12490	0.8828125	Overturned Decision of Health	A 36-year-old male underwent a CT angiography (CTA) of the heart due to trace
MN07-7166	0.8828125	Overturned Decision of Health	A 26-year-old woman with fibromyalgia and depression was prescribed Lyrica 15

Gold Layer: Purpose Built Views



Gold Layer: What are the Key Questions for a Decision Maker in the IMR Committee?

User Persona: Director, Dept of Insurance California



Role: Responsible for ensuring fairness and efficiency in medical insurance review process

Key Questions

- ③ Is our review fair and are outcomes consistent over time?
- ③ Where are the clinical hotspots with highest rates of overturned decisions?
- ③ Are there any demographic discrepancies in the review outcomes?

Key Question 1: Is our review fair and are outcomes consistent over time?

- How are overturn rates and communication sentiments evolving year-over-year?
- Are we seeing sustained changes in decision quality?

Logic

TABLE: IMR_YEAR_DETERMINATION

```
1 CREATE OR REPLACE TABLE GOLD.IMR_YEAR_DETERMINATION AS
2 SELECT
3   d.Report_Year,
4   r.Determination,
5   COUNT(*) AS Num_Reviews,
6   AVG(f.Sentiment) AS Avg_Sentiment
7 FROM SILVER.FACT_THR f
8 JOIN SILVER.DIM_DATE d ON f.Date_SK = d.Date_SK
9 JOIN SILVER.DIM_REVIEW r ON f.Review_SK = r.Review_SK
10 GROUP BY d.Report_Year, r.Determination;
```

Totaling all reviews

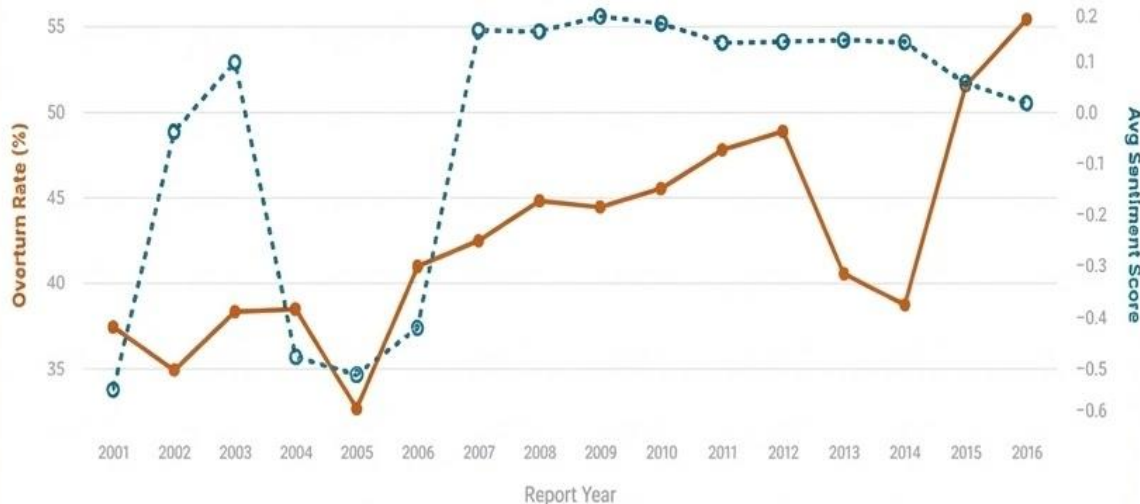
Calculating avg
communication tone

Aggregating to the
yearly level

Result

Overturn Rate & Sentiment Trends

Are we seeing rising friction (Negative Sentiment) or rising error rates (Overturn Rate)?



Key Question 2: Where are the clinical hotspots with highest rates of overturned decisions?

- Which combinations of diagnoses and treatments have the highest rates of being overturned?
- Where should we focus our clinical policy review?

Logic

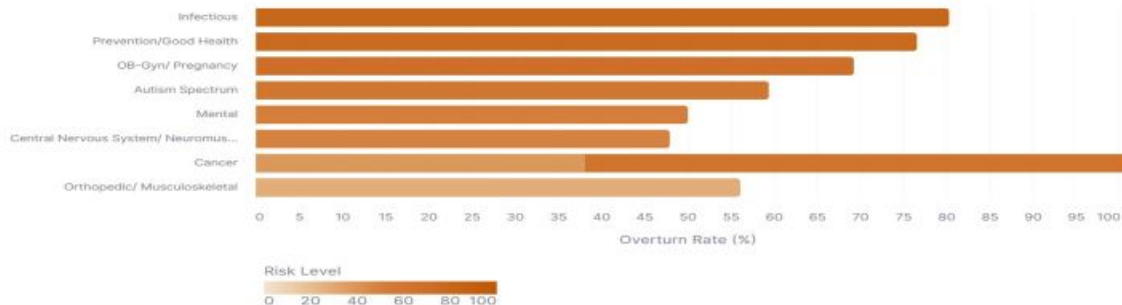
TABLE: IMR_DIAG_TREAT_OUTCOME

```
CREATE OR REPLACE TABLE GOLD.IMR_DIAG_TREAT_OUTCOME AS
SELECT
  dd.Diagnosis_Category,
  dt.Treatment_Category,
  r.Determination,
  COUNT(*) AS Num_Reviews
FROM SILVER.FACT_IMR f
JOIN SILVER.DIM_DIAGNOSIS dd ON f.Diagnosis_SK = dd.Diagnosis_SK
JOIN SILVER.DIM_TREATMENT dt ON f.Treatment_SK = dt.Treatment_SK
JOIN SILVER.DIM_REVIEW r ON f.Review_SK = r.Review_SK
GROUP BY dd.Diagnosis_Category, dt.Treatment_Category, r.Determination;
```

Result

Top Risk Categories by Overturn Rate

Which medical categories are being wrongfully denied most often? (Focus: >100 Cases)



Top Risk Combinations

Specific Diagnosis/Treatment pairs with highest reversal rates

Diagnosis	Treatment	Rate %
Infectious	Pharmacy/Prescription Drugs	80.3
Prevention/Good Health	Diagnostic Imaging, Screening and Testing	76.6
OB-Gyn/ Pregnancy	Diagnostic Imaging, Screening and Testing	69.3

Key Question 3: Are there any demographic discrepancies in the review outcomes?

- Are IMR outcomes equitable across different patient age groups?
- Can we proactively detect and investigate potential disparities?

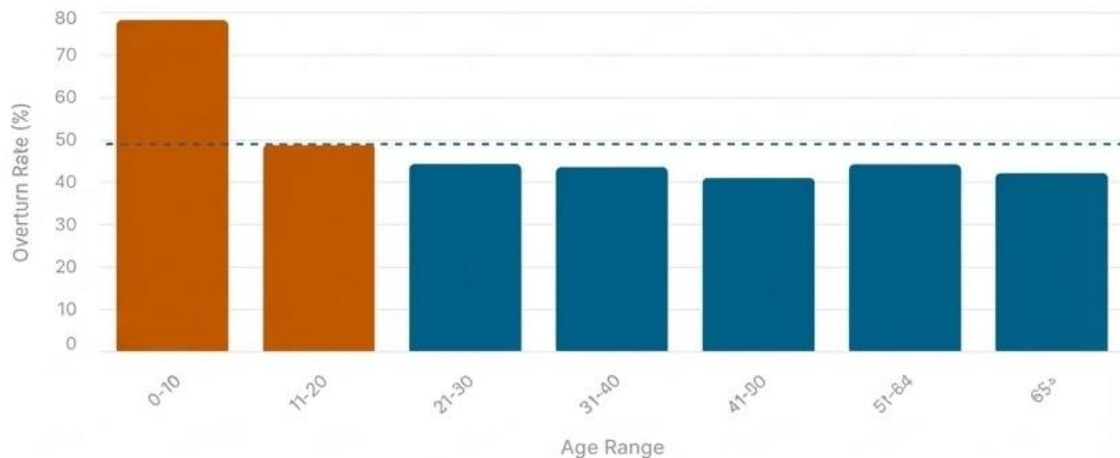
Logic

TABLE: IMR_DEMOGRAPHICS_OUTCOME

```
CREATE OR REPLACE TABLE GOLD.IMR_DEMOGRAPHICS_OUTCOME AS
SELECT
  p.Age_Range,
  p.Patient_Gender,
  r.Determination,
  COUNT(*) AS Num_Reviews
FROM SILVER.FACT_IMR f
JOIN SILVER.DIM_PATTENT p ON f.Patient_SK = p.Patient_SK
JOIN SILVER.DIM_REVIEW r ON f.Review_SK = r.Review_SK
GROUP BY p.Age_Range, p.Patient_Gender, r.Determination;
```

Result

Overturn Rate by Age Group



Equity Concern Detected

Significant variance of 37.2% between age groups. 0-10 patients experience 78.1% overturn rate vs. 41-50 patients at 40.9%. Further investigation recommended.

Engineering for Growth with Incremental Data Loads

Challenge: Data warehouses must handle new, incoming data without requiring a full reload. We simulated this by introducing a small file of 50 new IMR records.

Process: We implemented a mechanism to automatically detect and load this new data into the Bronze, Silver, and Gold layers, ensuring all tables remain synchronized.

Table_Name	Row_Count (Before)	Row_Count (After)	Change
BRONZE.IMR_RAW	19,245	19,295	+50
SILVER.FACT_IMR	19,245	19,295	+50
GOLD.IMR_YEAR_DETERMINATION	19,245	19,295	+50

	STATUS	TABLE_NAME	ROW_COUNT
0	After Load	FACT_IMR	19,295
1	After Load	DIM_DIAGNOSIS	394

	TABLE_NAME	TOTAL_REVIEWS_TRACKED
0	IMR_YEAR_DETERMINATION	19,295
1	IMR_DIAG_TREAT_OUTCOME	19,295
2	IMR_DEMOGRAPHICS_OUTCOME	19,295